News Shocks in Prediction Markets

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The question driving this thesis

Do news shocks about a company produce a measurable, predictable price reaction in that company's Polymarket contracts?

Everything that follows answers this question directly.

Context

- Data: 6,928 resolved contracts across 86 companies (Apr 2024-Aug 2026), matched against 546,657 GDELT articles
- No existing tool answers the question above directly, so this project builds two new methods
- GDELT only timestamps articles to the day, so every result below measures the 24-hour price reaction, the only window this feed supports

Method: the mechanics

Matching picks the right article for a contract, combining text similarity and trading-volume spikes:

$$\text{match_quality} = w_1 \cdot \text{logit}(\text{cosine similarity}) + w_2 \cdot z(\text{volume spike})$$

$$\text{match_confidence} = \text{sigmoid}(\text{match_quality})$$

Consistency forces a company's related contracts (above $150, $160, $170) to react in the same direction, weighted by volume, via isotonic regression (PAVA):

$$\min \sum_i v_i (y_i - r_i)^2 \quad \text{subject to} \quad y_1 \leq y_2 \leq \cdots \leq y_n$$

Method: why these, and why they should work

Both methods are new combinations of established tools, not inventions from scratch.

Combining independent signals in log-odds space is standard practice (the same math behind naive Bayes classifiers). Volume spikes as a marker of real news arriving is a standard idea in market microstructure research, going back to classic event-study methodology. Isotonic regression is a long-established tool for order-restricted inference (Barlow, Bartholomew, Bremner & Brunk, 1972), and the logic that related contracts should price consistently is the same logic behind no-arbitrage pricing.

What's new: applying both combinations to this specific problem, matching news to prediction-market contracts and enforcing no-arbitrage-style consistency across a strike ladder.

Result 1: contracts do react to news, with one caveat found since

Method	Spearman ρ	n
Matching score (text + volume)	**0.509**	14,717
Text similarity alone	0.233	14,717

This is half the answer to our question: a company's news and its market's trading behavior are measurably linked. Every later result depends on this step, it is how we built a trustworthy dataset of article-reaction pairs at all.

New finding (next section): part of the 0.509 number is inflated. Validated against the real stock's own reaction instead of Polymarket's own data, the matching score drops to ρ=0.029; text similarity alone holds up better (ρ=0.092). The volume component likely rode a coincidence specific to Polymarket's own data.

Result 2: the consistency correction makes predictions worse

Category	R^2 (raw)	R^2 (corrected)
Price-target	0.084	0.073
Valuation	0.746	0.363

We expected this to help: related contracts should price consistently, by no-arbitrage logic. The correction smooths away contract-specific variation the prediction models depend on, so we predict from the raw reaction.

Result 3: models land close to guessing the average

Category	Best model	R^2
All (pooled, n=5,151)	Linear	−0.010
Corporate event (n=3,157)	Linear	−0.021
Price-target (n=598)	LSTM	**+0.010**
Revenue-target (n=9)	TCN	−0.016
Valuation (small n)	Linear only	−0.255

This is the other half of the answer. If Polymarket prices already reflect public news quickly, this is exactly the pattern we'd expect: a reaction (Result 1) whose size depends on information beyond the article itself.

Result 4: one of four pre-registered predictions holds up

Prediction	Result
Richer embeddings beat simpler ones	Not supported
Best model differs by category	Not supported
Quantitative articles beat high-attention ones	Supported (n=28)
Best ladder category beats corporate events	Not supported

Result 5: Polymarket tracks the real stock's direction

Test	Spearman ρ	95% CI
Direction vs. direction	**0.119**	[0.047, 0.188]
Size vs. size	0.015	includes zero

CI from a company-clustered bootstrap. A company can show a huge real-world move alongside a tiny Polymarket reaction, or the reverse.

Result 6: a tradeable signal, for price-target contracts

$$\text{reaction} = \kappa \cdot D \cdot \frac{\text{stock return}}{\text{stock volatility}} + \delta$$

Term	Value
D (theoretical sensitivity)	1.702
κ (efficiency ratio)	0.093
δ (baseline drift)	−0.184

D is a known constant (Birnbaum 1968), not fit to our data: the slope a frictionless digital option would have. These contracts move at about 9% of that rate. δ is the 24h drift with no stock move, consistent with favorite-longshot overpricing correction (Snowberg & Wolfers 2010).

Result 6: validated and backtested

Check	Result
Correlation across train / val / test	0.18 / 0.14 / 0.18
Company-clustered bootstrap	95% CI [0.043, 0.288]
Backtest vs. 5,000 random permutations	100th percentile
Bonferroni-corrected across 23 models tried	p=0.0044

Mean P&L of +0.042 log-odds per trade, Sharpe ratio 14.3 (annualized). Keeps 98%+ of that profit under realistic trading costs (10-40bps), concentrated in a few large moves rather than spread evenly. Spans 16 companies, including 4 banks absent from training.

So, does Polymarket respond to news shocks?

Yes, and the picture sharpens as we look closer. Polymarket's own data shows a reaction (Result 1); every model we tested lands close to guessing the average (Results 2-4).

Checked against the real world, Polymarket's reaction tracks the real stock's direction (Result 5). For price-target contracts, that relationship strengthens into a tested, tradeable signal (Result 6).

Scoping decisions

- Market-cap contracts (4 markets): dropped, too few to draw a conclusion from
- Valuation contracts (110 markets): kept, below our 200-contract minimum sample size, but every other check passes
- Revenue contracts: given a separate data split so training was possible at all, instead of being excluded

Limitations

- GDELT gives day-precision timestamps, a structural limit of this feed
- `resolved_at` and `parent_event_id` are each about 40% backfilled from Polymarket's API; fallbacks cover the rest
- The revenue-target result rests on 19 training examples
- The price-target trading signal (Result 6) spans 16 companies; transaction costs come from published research rather than Polymarket's measured spreads

Next steps

- Measure Polymarket's actual historical bid-ask spread, to replace Result 6's cost estimate (currently from published research) with a real one
- Decide together how much weight Result 6 should carry given its narrow base (16 companies)
- Revisit the valuation-contract sample-size exception if more data becomes available